Shivaram Kumar Jagannathan

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Research Interests

Spatial-Semantic Models · Robot Perception · Long-Term Spatial Memory · SLAM & Localization · Semantics for Robot Reasoning · Machine Learning for Robotics

Education

Manipal Institute of Technology Bangalore, India
Bachelor of Technology in Information Technology, CGPA: 8.5/10 *Expected Graduation: 2026*

- Relevant Coursework: Modern Robotics, Embedded Systems, Computer Vision, Control Systems, Machine Learning, Kinematics and Dynamics

Publications & Research Output

Mani-Imagine: 3D Occlusion Aware Task and Motion Planning for Novel Object Search
Antareep Singha, Shivaram Kumar Jagannathan, Yoonwoo Kim, Yoonchang Sung
Target: ICRA 2027

VLM enabled Semantic Soft-Collision Manipulation
Shivaram Kumar Jagannathan, Yoonchang Sung
Bachelor's Thesis

Research Experience

Nanyang Technological University, Singapore Jan. 2026 - Present
Researcher *Singapore*

- Working at the Algorithmic Robotics Group, under Dr. Yoonchang Sung
- Co-authoring Mani-Imagine, a task and motion planning algorithm that leverages spatial 3d reconstruction to inform novel object search occluded in clutter.
- Built the complete e2e multi-object 3d reconstruction pipeline using SAM3D, with object localization using FoundationPose, pointcloud based grasp generation using CapGrasp and wrote a smooth torque control system for controlling the Franka FR3
- Core tech stack - SAM3D, FoundationStereo, FoundationPose, Qwen2.5 VL, PyBullet, LibFranka (C++), CLIP, CAPGrasp, GraspGen
- Authoring an algorithm paper on leveraging imagination using physics simulation to determine semantically safe manipulation in contact rich scenarios.
- Setup the infrastructure for the 3D Vision, control systems and planning for the robots in my laboratory.

Centre of Excellence in Autonomous Mobility, MIT Bangalore Jan. 2025 - Present
Technical Lead *Bangalore, India*

- Wrote a Unscented Kalman Filter to fuse IMU with GNSS readings accounting for RTK corrections
- Outfitted LIO-SAM to utillize multi-channel semantic scan context for improved loop closure and visual place recognition hypothesis generation, measured through recall@1.

IIT Bombay Robotics Lab Aug. 2024 - Jan. 2026
Hackathon Lead *Mumbai, India*

- Designed and implemented a warehouse automation project, where I led a team of 3 individuals
- Involved co-ordination of 2 robots, a UR5 that picked and placed packages from a shelf, onto an AMR platform, that delivered packages to specificied locations.
- Integrated MoveIt for motion planning for the UR5.
- Complemented autonomous delivery with a simple rover, hooked up to a 2D laser rangefinder and ultrasonics
- Implemented cartographer for mapping the warehouse, and used AMCL for localization

Technical Projects

AToM: Adaptive Teleoperation Middleware for Humanoid Manipulation Oct. 2025 – Oct. 2025

Lead Developer

- Developed a complete ROS2-based teleoperation framework enabling humanoid robot control through a Meta Quest VR headset and handheld controllers, mapping operator head and hand motion to robot neck and bilateral arm movement in real time
- Implemented a **multi-threaded teleoperation controller** that converted Meta Quest headset orientation and 6-DoF controller poses into robot motion commands using PyKDL, TF2, and adaptive motion scaling, achieving **≈20 ms end-to-end control latency**
- Designed a **VR-based Neck Tracking Node** that translated headset quaternion data into smooth neck trajectories through quaternion-to-Euler conversion, TF2-based pose transformations, trajectory clamping, and live camera feedback
- Integrated KDL-based **inverse kinematics solvers (LMA)** for bilateral arm and neck kinematic chains with reach projection, exponential smoothing (EMA), and retry-based convergence, achieving a **99% IK solve success rate**
- Developed adaptive reach projection and relative motion scaling techniques to preserve motion intent when VR controller targets exceeded the robot's kinematic workspace
- Built a **real-time joint trajectory execution pipeline** using ROS2 action interfaces and MultiThreadedExecutor, enabling deterministic low-latency control across concurrent tracking, IK, TF, and vision subsystems
- Integrated a **vision monitoring subsystem** using OpenCV and CvBridge for continuous image streaming, recording, and operator feedback during teleoperation sessions
- Focused on natural human–robot motion mirroring, constraint-aware control, and immersive VR teleoperation for humanoid manipulation and remote robotics research

Vision-Based Quadcopter for Crop Health Monitoring Aug. 2025 - Nov. 2025

Lead Developer

- Achieved **stable autonomous flight with ±0.1m positional accuracy** across all axes using attitude control and vision-based pose estimation, eliminating reliance on IMU or odometry sensors
- Designed and implemented a **Linear Quadratic Regulator (LQR)** controller with full-state feedback for optimal trajectory tracking, outperforming traditional PID approaches in stability and convergence speed
- Developed a complete **ROS2 action-based waypoint navigation system** with service-client architecture, enabling sequential autonomous flight through multiple 3D coordinates with real-time feedback
- Integrated **WhyCon visual marker tracking** for visual pose estimation.
- Implemented **adaptive stability monitoring** with sphere-constraint validation, requiring continuous presence within tolerance bounds and dual validation against both raw and filtered position data
- Developed an **RC command mixer** with thrust linearization, attitude control mapping, and real-time saturation handling to translate LQR control outputs into PWM commands for quadcopter actuators
- Implemented a crop-health identification system using YOLO by using the on-board camera.
- Demonstrated vision-only autonomous navigation without traditional sensor fusion, proving viability of camera-based control for resource-constrained aerial platforms

Grants & Awards

IIT Bombay eYantra Robotics Competition Aug. 2024 - Jan. 2025

Lead Researcher - Top 1% among participants *Mumbai, India*

- Led a team to develop advanced autonomous navigation algorithms with real-time SLAM implementation

AWS DeepRacer Scholarship Jul. 2022 - Dec. 2022

Excellence in Reinforcement Learning *Bangalore, India*

- Recognized for excellence in reinforcement learning for self-driving models

Professional Experience

Virya Autonomous Technologies Jun. 2025 – Dec. 2025

Robotics Systems Engineer Intern *Bangalore, India*

- Developed a **global localization system** for autonomous industrial mobile robots (AMRs), achieving **0.05 m Absolute Trajectory Error (ATE)** against GNSS ground truth and an average **0.006 m Relative Pose Error (RPE)** across evaluation sequences.

- Implemented a **GNSS-based fallback relocalization mechanism** using the **CUAV UM982 RTK chipset**, automatically engaging when LiDAR and camera localization pipelines failed, leveraging RTK fixes for centimeter-level global accuracy.
- Designed a **feature detection and relocalization pipeline** leveraging **Visual Bag-of-Words (BoW)**, **Bayesian hypothesis generation**, and **TF-IDF likelihood scoring** for robust feature association and graph node matching within the SLAM backend.
- Introduced a **safety-critical covariance confidence metric**, published at high frequency, to quantify localization certainty across the entire SLAM chain (*feature extraction → hypothesis generation → likelihood ratio → feature matching → 6-DoF coarse alignment → ICP refinement → graph optimization*); enabling low-level controllers to autonomously trigger emergency stops or recovery maneuvers under uncertain localization states.
- Implemented an **Optical Flow–based odometry module** as a fallback to feature-based matching, ensuring reliable motion estimation under low-light, high-glare, or texture-deprived visual conditions.
- Advanced the perception pipeline by fusing data from the **Ouster OS-1 (128-channel) LiDAR** and **Intel RealSense RGB-D camera** via reprojection-based sensor fusion and extrinsic calibration—replacing the legacy 16-channel LiDAR system for enhanced spatial fidelity.
- Parallelized the **DLIO odometry engine** using **OpenMP** to exploit multi-core CPU architectures, achieving real-time SLAM updates and reduced odometry latency.
- Evaluated and benchmarked both **G2O** and **GTSAM iSAM2** as core optimization backends, analyzing convergence behavior, runtime performance, and incremental pose-graph update characteristics for large-scale maps.
- Integrated **Vertigo** for robust graph optimization to dynamically handle outlier loop closures and inconsistent spatial constraints, replacing static error thresholds with adaptive switchable factors for stable long-term mapping.
- Performed a comprehensive comparison of modern **SLAM frameworks** (Cartographer, FAST-LIO, LIO-SAM, RTAB-Map) under noisy industrial environments with frequent sensor outages, dynamic obstacles, and spurious LiDAR reflections.
- Extended RTAB-Map's modular framework to support heterogeneous odometry and sensor sources, integrating visualization and diagnostic telemetry in **Foxglove Studio**.
- Conducted **multi-session SLAM experiments** spanning **over 6 km of robot trajectories**, generating merged maps covering **approximately 3 km^2** through cross-session loop closures and global pose-graph alignment.
- Developed an internal **environment-authoring toolkit** enabling large-scale map visualization, lane and safety-zone definition, behavior annotation (docking, charging, pausing), and automatic generation of optimized waypoint graphs functioning as a global planner for autonomous robots.
- Engineered a high-performance **point-cloud rendering subsystem** using OpenGL optimizations, achieving **loading and visualization of 30 million points in under 5 seconds** to support large-scale environment inspection and workflow design.
- Integrated perception, planning, and visualization modules into a unified, extensible interface allowing operators to author navigation pipelines directly from mapped environments, significantly improving tooling efficiency and deployment workflows.

Exponent Energy Jun. 2023 - Aug. 2023
Software Engineer Intern *Bangalore, India*

- Deployed a kubernetes cluster on AWS for a charging station locator, for 100+ charging station located in Bangalore.
- Wrote a CAN-based communication reliability and fault detection firmware for the electrical power systems, publishing data to the cloud.

References

Dr. Yoonchang Sung Nanyang Technological University
Assistant Professor, Algorithmic Robotics Group

Dr. Sravani V Manipal Institute of Technology
Faculty Advisor, Senior Scale, CoE

Arun Vinayak Exponent Energy
CEO

Athar Ahmed Virya Autonomous Technologies
Robotics Systems Team

Sabarish Gurusubramaniam Virya Autonomous Technologies
CTO